João André **Roque Costa**

MSc Graduate, Computer Science and Engineering

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Early-career data scientist with an MSc in Computer Science and Engineering and hands-on experience applying **Python** to machine learning, data analysis, and **LLM** prototypes. Built a Claude SDK and Azure AI Foundry data-migration prototype, prototyped HR-report automation, and developed a parallelizable method for nonlinear dimensionality reduction. Brings clear technical communication from teaching and cross-functional project work, plus Databricks certification in Generative AI Application Deployment and Monitoring.

Professional Experience

Deloitte — *Coimbra, Portugal*

Tech Analyst — *Apr. 2026 - Oct. 2026*

- **Built** a cross-functional **LLM** prototype for data migration and synthetic data generation using Claude SDK, Azure AI Foundry, Azure Blob Storage, and MongoDB.
- **Prototyped** HR report generation with Python and Playwright, and contributed technical design for a client proposal using RAG and OCR to automate document workflows.
- **Delivered** an Azure subscription migration for Databricks and Azure Data Factory assets within the planned timeframe, coordinating Managed Instances and dependent services.

Instituto Superior Técnico — *Lisbon, Portugal*

Teaching Assistant — *Oct. 2024 - Jul. 2025*

- **Taught** and evaluated practical classes in Databases and Data Science, tracking student progress and explaining technical concepts to learners with varied experience.
- **Guided** Python projects in data treatment, analysis, and visualization using pandas and matplotlib, providing constructive feedback and adapting support to each student.

Education

Instituto Superior Técnico — *Lisbon, Portugal*

M.Sc. Computer Science and Engineering — *Sep 2023 - Nov 2025*

- Focused on **Artificial Intelligence** and data science, applying machine-learning foundations to modeling and data-driven problem solving.
- **Relevant Coursework**: Deep Learning, Data Science, Autonomous Agents and Multi-Agent Systems, Optimization Algorithms.
- Thesis: **Maximum Variance Unfolding on Disjoint Manifolds (MVU-DM)**.
  - Developed a decomposition-based extension to MVU for **nonlinear dimensionality reduction** on disconnected datasets.
  - Designed a parallelizable method that improved scalability on large datasets while preserving local geometry and producing embeddings comparable to standard MVU.

Instituto Superior Técnico — *Lisbon, Portugal*

B.Sc. Computer Science and Engineering — *Sep 2020 - Jul 2023*

- Studied algorithms, distributed systems, operating systems, and database theory, building foundations for maintainable software and data-intensive systems.
- Applied **C/C++** in systems projects with emphasis on memory management, and designed relational schemas in **SQL**.

Extracurricular

Personal Software Projects (Self-Hosted)

- Developed a Wallapop price tracker that monitors user-defined product baskets and alerts when total prices fall below a target or items are substantially discounted.
- Built a Discord **ELO matchmaking system** that queues players by rating and stores match history in PostgreSQL, using Redis for caching and queue management.

Skills & Competencies

Data Science	Machine learning, deep learning, data analysis, data visualization, nonlinear dimensionality reduction
Programming	**Python** (pandas, NumPy, scikit-learn, TensorFlow, Keras, matplotlib, Playwright), C/C++
LLM & Cloud	LLM prototyping with Claude SDK, Azure AI Foundry, Azure, Databricks, Azure Data Factory, Azure Blob Storage, Azure Managed Instances; RAG/OCR proposal design
Data & Databases	**SQL**, PostgreSQL, MongoDB, MySQL, Redis, Microsoft SQL Server (SSMS), pgAdmin
Software Systems	Git, Docker, Linux, Bash scripting, REST APIs, Python automation with Playwright
Languages	Portuguese (Native), English (Fluent)
Collaboration	Cross-functional stakeholder communication, mentoring, technical presentation
Additional	LaTeX, ArchiMate